

—MASTER'S COMPREHENSIVE EXAMINATION

Part II Applied (2 hours) December 1st, 2018.

**Instructions: Attempt any three out of the following four questions on both pages.
Mention which three you would like to have graded. Open book and open notes.**

1. A random sample of 20 adult Busch Bears is obtained from a forest in New Jersey. The following stem & leaf plot gives the weights in pounds for these animals

1		2789
2		12346679
3		1358
4		0112

- a) Compute the median weight in the sample_____
- b) Compute the interquartile range of weight in the sample_____
- c) Test the null hypothesis that the median weight of an adult Busch bear is heavier than 39 pounds against the alternative that it is not heavier than 39 pounds at $P = 0.001$.
- d) A scientist believes that 25% of adult Busch Bears weigh 11 pounds or less. Is this hypothesis refuted by the data (explain your answer)?

2. A market survey research group took a random sample of 800 teenage boys, and an independent random sample of 800 teenage girls from all teenagers in the United States. Each teenager was asked the following two questions:

Question 1: Do you use Facebook? Circle one: Yes or No

Question 2: Do you use Snapchat ? Circle one: Yes or No

Every teenager answered either Yes or No.

On Question 1: 392 boys, and 424 girls answered Yes.

On Question 2: 536 boys and 576 girls answered Yes.

- a. Based on this data is there statistically significant evidence (at 0.05 level) of a difference between the proportion of U.S. teenage boys and the proportion of U.S. teenage girls who use Facebook? Carry out the test.

b. Your Director of Market Research asks you “Based on this data, is there statistically significant evidence (at 0.05 level) of a difference between the proportion of U.S. teenage boys who use Snapchat, and the proportion of U.S. teenage boys who use Facebook.?” You realize that here the two sample proportions are not independent. Some of the 800 boys may have answered No to both questions, some may have answered Yes to both questions, and so on, but this data is not available. However, you do know from the above data that the number of boys who answered No to both questions is between 0 and 264. Hint: try 0, try 264, try other possible values x , to see if the conclusion is the same for any values of x the range from 0 to 264.

3. A golf ball manufacturer has developed a new golf ball that the manufacturer claims will lead to longer drives on tee off. The manufacturer has an experiment conducted by an independent testing company. The testing company draws independent random samples of 10 high school team golfers, 10 college team golfers, 10 professional golfers. Each golfer does 20 drives with the new ball, and 20 drives with the leading competitor’s ball (the order of the 20 drives is randomized for each golfer). The response variable is the distance of the drive. Assume that the response variable is normally distributed for a given golfer using a given ball, and that the assumptions of the Analysis of Variance apply. The main interest is in effect on the response variable of the BALL (new vs competitor). The manufacturer is also interested in whether the BALL effect differs for different LEVELS of golfers (Highschool, College, Professional).

a. Write out the ANOVA table showing Source of Variability, Degrees of Freedom. Note that your Sources of Variability should include the variability of length of DRIVES for a given GOLFER at a given LEVEL using a given BALL;(This Source of Variability can be listed as ERROR). In the ANOVA table you should indicate that factor A_1 is

nested within A_2 which is nested within (A_3 crossed with A_4) by using the notation

$A_1|A_2|A_3A_4$.

- b. For each Source of Variability, state whether effect is Fixed or Random, and write out Expected Mean Squares,
- c. Use the Expected Means Squares in (b) to find the Sources of variability in denominator of F ratios to test the significance of the ANOVA Sources of Variability
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4. Puntious Pilot the Roman who pioneered kicking egg-shaped air inflated balls is building a linear model for force applied to the ball and the distance the ball travels. The model is $Y = bX + \varepsilon$ where X is the force Applied, Y is the distance the ball travels and ε is iid error with mean 0 and constant variance σ^2 . **NOTE that the intercept of this model is 0.** Puntious has the following data to use to estimate b

X	1	2	3	4	5	6	7
Y	3	4	5	9	8	20	18

- a. Write out the design matrix X
- b. Estimate b
- c. Estimate σ^2
- d. Estimate the variance of the estimate for b you gave above
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